Project Name

Project ID

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT (Maximum 300 words)

Sri Lanka frequently experiences natural disasters such as floods, landslides, and cyclones. During these events, many affected people rely on SMS or messaging platforms to request help, especially when voice calls are not possible due to weak network connectivity. These messages are usually very short and often written in a stressful situation. As a result, they may contain spelling mistakes, abbreviations, emojis, and mixed languages such as Sinhala or Tamil written using English letters. Because of this, it becomes difficult for emergency operators to quickly understand the message and determine how urgent the situation is.

This research focuses on developing an **AI-based low-bandwidth text processing system** that can analyse emergency SMS and WhatsApp messages during disaster situations. The system will identify the language of the message, clean and normalize the text, and extract important details such as the type of incident, possible location information, number of people affected, and the type of help required. It will also classify the urgency level of each message into four categories: Critical, High, Medium, and Low. In cases where the message does not provide enough information, an AI-based SMS agent will ask a small set of follow-up questions to gather additional details from the sender.

The extracted information will then be converted into structured JSON data that can be used by emergency response dashboards or decision-support systems. To ensure the system performs well in real-world conditions, the models will be trained and tested using a combination of simulated disaster messages and synthetic datasets that include noisy and incomplete text.

The expected outcome of this research is to improve how emergency text messages are understood and prioritized during disasters. By helping operators quickly identify critical situations from short and noisy messages, the system can support faster emergency response and more effective disaster management.

Keywords: Low-Bandwidth Emergency Communication, SMS Processing, Multilingual NLP, Urgency Classification, Disaster Response

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
NLP	Natural Language Processing
ML	Machine Learning
LLM	Large Language Model
NER	Named Entity Recognition
JSON	JavaScript Object Notation
SMS	Short Message Service
API	Application Programming Interface
SDG	Sustainable Development Goals
DMC	Disaster Management Centre

1. INTRODUCTION

1.1 Background and Context

Sri Lanka frequently experiences natural disasters such as floods, landslides, cyclones, and severe storms. During such events, emergency communication channels—particularly SMS hotlines and messaging platforms like WhatsApp—receive a sudden surge of distress messages from affected communities. These messages are often very short, fragmented, and written under stressful conditions, as people attempt to request help using limited connectivity and mobile devices.

In many cases, disaster messages contain informal language, abbreviations, spelling errors, emojis, and code-mixed text, where Sinhala or Tamil is written using English characters. Poor network conditions may also cause truncated or delayed messages, making it difficult for emergency operators to clearly understand the situation. As a result, extracting important information such as the type of incident, location, number of affected people, and urgency level becomes challenging.

Emergency hotline operators and disaster management authorities rely on this information to make rapid decisions and deploy appropriate rescue resources. However, manually interpreting large volumes of unstructured SMS messages during disasters can significantly increase response time and the risk of misinterpreting critical information.

Existing emergency communication systems are typically designed for structured inputs and high-resource languages, and they often struggle to process short, noisy, or multilingual text messages in real time. Therefore, there is a need for an AI-powered low-bandwidth text intelligence system capable of analysing emergency SMS messages, extracting key information, determining urgency levels, and converting them into structured intelligence that can support faster and more accurate disaster response decisions.

The proposed solution is an AI-assisted emergency triage system that includes human review. It processes voice calls and low-bandwidth text messages.

It has several components:

(1) multilingual voice recognition and speech-to-text transcription

(2) low-bandwidth SMS and WhatsApp message processing

(3) urgency classification and emotional distress detection

(4) structured data output to an operator dashboard for quick decision-making.

By merging multimodal AI processing with human oversight, the system aims to cut response times, improve the prioritization of life-threatening cases, and help operators with resource allocation.

This research supports the United Nations Sustainable Development Goals (SDGs), especially SDG 3 (Good Health and Well-being), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 11 (Sustainable Cities and Communities). It does so by improving emergency responsiveness and public safety. It also contributes to the AI and Data Science research area, using modern natural language processing, speech recognition, and machine learning techniques to tackle disaster management challenges.

Literature Review

This section reviews recent scholarly work on emergency triage systems, natural language processing (NLP), and speech/text analysis in emergency contexts. It highlights existing research approaches, products/services, and limitations that inform the design of the proposed **AI-Based Emergency Call and SMS Triage System** for Sri Lankan disasters.

Scholarly Research Approaches

1. **AI Models for Triage Prediction in Emergency Departments**
   Lan siaux *et al.* compared three AI models (NLP, LLM, and JEPA) for predicting triage outcomes in emergency departments. Their results showed that large language model approaches like LLMs outperformed traditional NLP and other models, underscoring AI's potential in structured medical priority prediction but within well-defined clinical datasets rather than unstructured verbal or SMS input.

2. **Semantic Parsing and Multilingual Triage**
   Systems like TriagE-NLU integrate semantic parsing with multilingual triage in telemedicine, generating structured outputs and intervention classifications. They use synthetic datasets and federated learning to handle multilingual inputs more robustly than simple classifiers.

3. **Clinical Text NLP for Emergency Surveillance**
   Studies applying transformer-based NLP to triage texts demonstrated that language models could classify clinical presentations effectively, even with limited data, emphasizing the importance of augmented training data for performance in real-world settings.

Emergency NLP & Speech Systems in Practice

1. **Medical Decision Support with NLP & Machine Learning**
   Recent work shows combining structured numerical and unstructured textual data can create objective, AI-supported emergency triage models. Intelligences how emergency intelligence systems can benefit from deeper text analysis rather than relying solely on human judgment.

2. **NLP for Disaster Response & Crisis Management**
   Broader research on NLP in disaster response highlights automated classification and

summarization of crisis data, including social media and emergency communications, underscoring the growing importance of real-time NLP for situational awareness.

Existing Products & Services

1. **Commercial & Open-Source Speech-to-Text in Emergency Settings**
   Benchmarking studies on ASR show transformer-based systems (e.g., Whisper) maintain meaning understanding in noisy environments — a crucial requirement for emergency call transcripts. However, even top systems degrade significantly under very low signal-to-noise ratios.

Existing products like commercial 911 triage tools assist operators with structured data but often rely on preformatted inputs or supported languages. These systems typically lack adaptability to short, fragmented SMS content or code-mixed languages, and they may not integrate low-bandwidth messaging channels.

Limitations of Current Systems

- **Language Diversity and Low-Resource Support:**
  Many NLP and ASR systems are optimized for high-resource languages (e.g., English, Spanish), with performance dropping sharply for under-resourced or code-mixed language inputs.
- **Unstructured Input Challenges:**
  Most current emergency triage models focus on structured or long-form text (e.g., clinical symptoms or patient records) rather than short, noisy messages typical of disaster SMS.
- **Integration of Multimodal Inputs:**
  Few systems unify voice and text channels, instead addressing them separately, limiting real-time comprehensive triage intelligence.
- **Operational Deployment Gaps:**
  Despite promising research results, there is limited evidence of deployment at scale, particularly in low-resource or multilingual emergency communication environments.

1.3 Research Gap

Despite incredible strides made in AI-supported crisis situations, there remains a major shortfall in identifying the most optimal method integrated to the crisis management environment of Sri Lanka.

1. What is Missing

Most emergency triage systems mainly focus on the majority of resources and operate primarily in English or provide the majority of usability on structured data files and then will rarely or seldom be able to utilize SMS text, WhatsApp or lesser bandwidth (more noise) channels of communication [1]. The majority of the research and commercially available options currently cannot accept either a code mix (Singlish and Tanglish) or Romanized Sinhala/Tamil. There are limited capabilities of integrating voice and text together through multiple modes so operators manually must operate based on separate channel information, resulting in slower overall emergency response.

2. Current Challenges to Improving Current Techniques

(i) Spoken and Written Language Limitations: ASR (Automatic Speech Recognition) and NLP (Natural Language Processing) models were developed using data from English (a language that may be considered to be a high-resource language). As such, accuracy will likely decrease when these models are applied to low-resource languages or code-mixed messages. [1]

(ii) Short and Noisy Messages: Triage systems generally use structured long-form textual formats for processing messages that are often only partially complete or are disrupted (ei. short messages). The majority of current triage systems may not have the ability to handle short messages adequately. [2]

(iii) Lack of Multi-Modal Fusion of Voice and Text: An evidence-based approach does not integrate voice and text input in real-time; therefore, the timeliness of the decision process is limited. [1]

(iv) Operational Constraints: Many current triage systems have been developed to function with stable and predictable network conditions and/or high-level bandwidth facilities. The availability of such resources on the ground during disasters in Sri Lanka is not assured. [3]

3. Rationale for Innovation & Creativity

The innovation that will fill identified voids is a "multilingual, multi-modal, artificial intelligence-based emergency triage system" specifically designed to be utilized during the many types of disasters that occur in Sri Lanka. This proposal offers creative solutions by:

1.Simulated processing of low-bandwidth text messages along with live SMS calls in real time.

2.Being able to process code-mixed and Romanized languages common in the Sri Lankan context.

4. Evidence-Based Argument

• Several research papers support the assertion that the application of transformer-based NLP models improves the classification of emergency text, [2] [4] but the performance level declines significantly for the classification of short, fragmented messages or low-resource type languages (Brooke et al., 2024).

Research on ASR in emergency settings demonstrates that prosodic features improve urgency detection in voice calls, yet code-mixed speech is rarely supported (Lansiaux et al., 2025).

Commercial 911/112 systems fail to integrate SMS and voice effectively, showing that a unified pipeline can reduce operator workload and improve response times.

By addressing these limitations, the proposed system introduces a first-of-its-kind solution in Sri Lanka that combines multilingual text and voice intelligence, low-bandwidth optimization, and human-in-the-loop decision support, offering both practical and socially impactful innovation.

2.1 Main Objective

To design, develop, and evaluate an AI-powered low-bandwidth text intelligence module capable of processing short, noisy, and multilingual SMS/WhatsApp messages during disaster situations and converting them into structured emergency intelligence with urgency prioritization for real-time decision support.

2.2 Specific Objectives

- **Short Message Language Detection**
  Develop a lightweight NLP module capable of detecting Sinhala, Tamil, English, and code-mixed text from very short SMS messages (3–5 words) with high accuracy.
- **Text Normalization for Noisy Messages**
  Design preprocessing techniques to handle abbreviations, emojis, repeated characters, and Romanized Sinhala/Tamil text to improve the interpretation of informal emergency messages.
- **Incident and Entity Extraction**
  Develop an extraction module to identify key disaster information from SMS messages, including incident type, location clues, number of affected people, and required resources.
- **Urgency Classification**
  Implement a hybrid rule-based and machine learning model that classifies messages into four urgency levels (Critical, High, Medium, Low) to support rapid disaster response prioritization.
- **AI-Based Follow-Up Question Agent**
  Design an AI SMS interaction agent that asks a small set of critical follow-up questions to collect missing information when the original message is incomplete.
- **Structured Intelligence Generation**
  Convert processed SMS information into structured JSON outputs that can be integrated with emergency dashboards and multimodal disaster intelligence systems.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach combined with experimental evaluation.

Design Science Research is appropriate because the study focuses on designing and implementing a novel technical artifact—a low-bandwidth emergency intelligence system capable of processing noisy SMS messages and generating structured disaster information. The artifact includes a text processing pipeline and an AI-based SMS interaction agent designed specifically for disaster communication environments with unreliable connectivity.

The study will involve:

- Designing the SMS intelligence pipeline
- Developing normalization and extraction models for short messages
- Implementing an AI agent that performs structured follow-up questioning
- Experimentally evaluating system performance using simulated and real SMS datasets

The research aims to evaluate how effectively the system can interpret incomplete, noisy, and low-bandwidth disaster messages and convert them into actionable intelligence for emergency response dashboards.

3.2 Technical Methods and Algorithms

The system consists of five main processing modules.

1. Language and Script Detection

Short-message language detection will be implemented using lightweight Natural Language Processing techniques capable of handling short text.

Methods include:

- Character n-gram based language detection
- Script identification for Sinhala, Tamil, and Romanized text
- Handling code-mixed messages (e.g., Sinhala written in English characters)

This step ensures that the system correctly identifies the linguistic context before applying normalization and extraction.

2. Text Normalization and Noise Reduction

Disaster messages often contain informal, abbreviated, or corrupted text. A preprocessing pipeline will normalize these messages using:

- Rule-based abbreviation expansion
- Emoji interpretation for emergency signals
- Removal of repeated characters (e.g., “helpppp”)

• Romanized Sinhala/Tamil conversion to standardized forms
• Spelling correction using dictionary matching

This normalization step improves the accuracy of downstream NLP models.

3. Incident and Entity Extraction

After normalization, the system extracts critical disaster information from SMS messages using **Named Entity Recognition (NER)** and rule-based pattern matching.

Entities extracted include:

• Incident type (flood, landslide, fire, medical emergency)
• Location references
• Number of people involved
• Resource needs (boats, ambulances, rescue teams)

Machine learning models such as **transformer-based NLP models or lightweight classifiers** will be used alongside rule-based extraction to handle short messages effectively.

4. Urgency Detection and Classification

Urgency classification identifies how critical a message is for emergency response prioritization.

The system classifies messages into four levels:

• Critical
• High
• Medium
• Low

Classification will be performed using **supervised machine learning models**, trained on labelled disaster SMS datasets. Features used include:

• Emergency keywords
• Linguistic patterns
• Message sentiment
• Message structure and brevity

This module enables rapid prioritization for emergency responders.

5. AI Agent for Interactive SMS Questioning

To compensate for incomplete or ambiguous messages, the system includes an **AI-based SMS interaction agent** that automatically asks a small set of structured follow-up questions.

The agent uses a **predefined decision logic** to trigger up to **five critical follow-up questions**, such as:

• What type of emergency is happening?
• How many people are affected?

• What is your nearest landmark or location?
• What help is needed?
• Are people injured or trapped?

The responses collected are combined with the original SMS message to **improve entity extraction and urgency assessment**.

This interaction design ensures the system can gather **more reliable situational information even when the first message is incomplete**.

3.3 Tools and Platforms with Justification

Several technologies will be used to implement the system.

Tool / Platform	Purpose	Justification
Python	Core development	Strong NLP ecosystem
Hugging Face Transformers	NLP modeling	Effective for text classification and entity extraction
spaCy	Named entity recognition	Lightweight and efficient
Twilio / SMS Gateway	SMS interaction	Enables real-time SMS communication
Firebase / Cloud Database	Data storage	Real-time updates for dashboards
FastAPI	Backend API	Efficient integration with other components
Docker	System deployment	Ensures portability and reproducibility

Table 1: Tools and purpose

3.4 Data Collection and Ethical Considerations

The research will utilize **both real and synthetic datasets**.

Data Sources

1. Public disaster message datasets
2. Synthetic SMS datasets generated through **noise simulation techniques**, including:
   • message truncation
   • spelling errors
   • code-mixed language
   • informal abbreviations

Synthetic data generation helps simulate **low-bandwidth disaster communication conditions**.

Ethical Considerations

The research will ensure:

• No personally identifiable information (PII) is stored
• Sensitive location information is anonymized

• Data is used only for research purposes
• Compliance with institutional research ethics guidelines

All datasets will be **securely stored and access-controlled**.

3.5 Validation Strategy and Performance Metrics

The system will be evaluated through **experimental testing and usability validation**.

1. Experimental Evaluation

Performance will be measured using:

• Language detection accuracy
• Entity extraction F1-score
• Urgency classification accuracy
• Precision and recall for incident detection

2. Robustness Testing for Noisy Messages

The system will be tested with:

• truncated SMS messages
• messages with spelling errors
• mixed-language messages
• incomplete disaster reports

This evaluation measures how well the system performs in **realistic low-bandwidth communication scenarios**.

3. Interaction Agent Evaluation

The AI SMS agent will be evaluated based on:

• improvement in extracted information after follow-up questions
• response completion rate
• accuracy improvement in urgency classification

4. System-Level Evaluation

Finally, the overall system will be evaluated based on:

• decision-support usefulness for emergency responders
• integration effectiveness with the emergency dashboard
• processing speed for real-time disaster intelligence

4. HIGH-LEVEL SYSTEM ARCHITECTURE

1. Overview of Proposed Solution

The proposed system is an **AI-based multilingual emergency triage platform** designed specifically for Sri Lankan disaster management. The solution processes voice calls and SMS/text messages in Sinhala, Tamil, English, and code-mixed variants (Singlish/Tanglish), transforming raw emergency communications into structured, prioritized intelligence for human operators.

The architecture follows a **modular pipeline design** with four core components, each handling a distinct phase of emergency communication processing. This decoupled design enables independent development, testing, and future enhancement of each component while maintaining clear integration interfaces.

Component	Name	Primary Function
C1	Voice Processing	Processes voice calls; performs language ID, ASR transcription, prosody analysis, incident extraction, and urgency classification
C2	AI Clarification Agent	Interactively engages callers to gather missing information using panic-adapted questioning and dynamic adaptation
C3	SMS/Text Processing	Processes SMS and text messages; handles language/script detection, entity extraction, and urgency classification, AI bot reply back to back to the messages
C4	Incident Fusion & Dashboard	Integrates outputs from C1/C2/C3; performs incident clustering, timeline reconstruction, conflict resolution, duplicate filtering; displays operator dashboard

Table 2: Component and function table

2. System Architecture Diagram

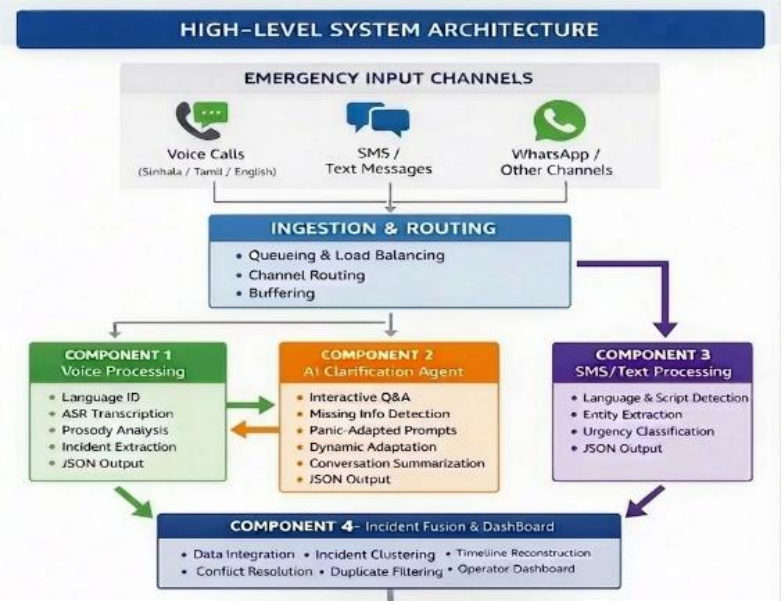


Figure 1: High Level System Architecture

4. Component Interactions

4.1 Data Flow Between Components

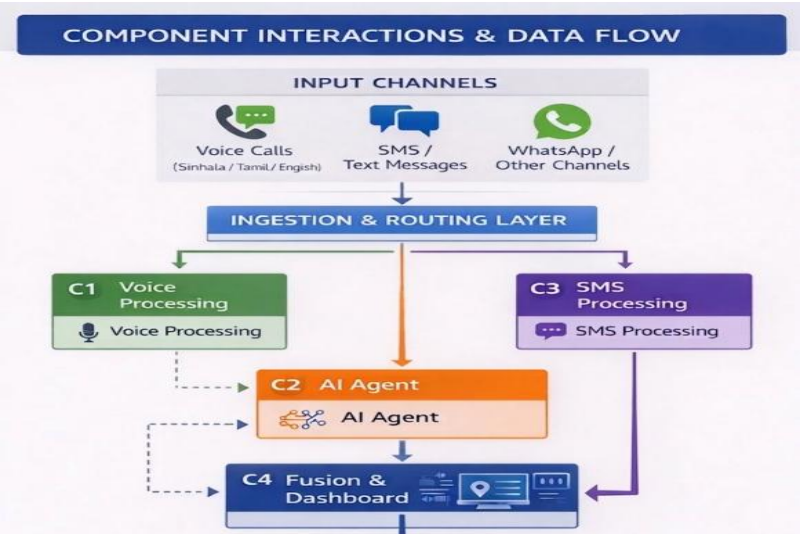


Figure 2: Data Flow Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Emergency Hotline Operators – Primary users who will receive AI-processed summaries of voice and text messages, enabling faster and accurate triage.

Disaster Management Authorities – Utilize system insights for resource allocation and strategic decision-making during disasters.

First Responders (Medical, Fire, Rescue Teams) – Receive timely alerts and instructions based on AI-prioritized emergency cases.

Civilians / Disaster-Affected Individuals – Indirect beneficiaries, as they receive faster and more effective emergency response.

System Administrators – Maintain system performance, manage datasets, and ensure proper functioning of the dashboard and AI modules.

5.2 Functional Requirements

The system shall process multilingual voice calls (Sinhala, Tamil, English, and code-mixed forms) and convert them into text with at least 90% accuracy.

The system shall process short, low-bandwidth SMS/WhatsApp messages, normalize text, and extract incident details, location, and urgency.

The system shall classify urgency levels as Critical, High, Medium, or Low for both voice and text inputs.

The system shall responds to massagers back to back using AI bot

The system shall integrate voice and text data to generate structured JSON outputs for operator dashboards.

The system shall provide human-in-the-loop functionality, allowing operators to review and modify AI-generated summaries and alerts.

The system shall store historical communications and incident data for audit, analysis, and training purposes.

The system shall generate real-time notifications to disaster management authorities and first responders based on priority.

The system shall support low-bandwidth and low-connectivity operation, ensuring functionality during disaster conditions.

5.3 Non-Functional Requirements

- *Security:*

  Ensure data encryption for storage and transmission.

  Implement access control to restrict sensitive data to authorized personnel.

- *Performance:*

  Process incoming voice/text data and display results on the dashboard within 5 seconds.

  Handle high-volume spikes during disaster events without system failure.

- *Usability:*

  Provide an intuitive operator dashboard with clear alerts and summaries.

  Support simple overrides and manual input corrections for human-in-the-loop validation.

- *Reliability:*

  Maintain system uptime of 99% during emergencies.

  Gracefully handle message truncation, out-of-order delivery, or repeated messages.

- *Scalability:*

  Accommodate increased communication load during large-scale disasters.

  Support addition of new languages or communication channels in the future.

- *Maintainability:*

  Modular system architecture for easy updates, debugging, and integration of new AI models.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target customer is the Disaster Management Centre (DMC) Sri Lanka — the national government body responsible for coordinating disaster preparedness and response across all 25 districts. The DMC has operationally confirmed the need for a system of this type. Secondary target customers include the Ministry of Public Security, Police Emergency 119, Provincial Disaster Management Units, and Colombo Municipal Council emergency services. The longer-term market includes SAARC-region countries with similar low-resource language emergency communication challenges — Nepal, Bangladesh, and Myanmar — and international humanitarian organizations (UNDP, Red Cross) operating in multilingual disaster contexts.

6.2 Value Proposition

- First and only AI emergency triage system supporting Sinhala, Tamil, and code-mixed Sri Lankan communication
- Reduces operator workload by over 40% by automating report triage, deduplication, and prioritization
- Reduces critical incident missed-response risk through false-negative-optimized urgency detection
- Enables operators to manage 3–5x more simultaneous incident reports without additional staffing
- Works on low bandwidth and SMS — critical for disaster contexts where voice networks may be saturated

6.3 Revenue Model

Stream	Description	Timing
Government License	Annual license fee for DMC, Ministry of Public Security, Police 119	Year 2 post-project
Implementation Services	One-time setup, integration, and operator training per agency	Year 2+
SaaS Subscription	Per-district or per-seat subscription for municipal councils	Year 3+

Regional Export License	White-label or adapted deployment for SAARC countries	Year 4+
Grant Funding	NSF Sri Lanka, ICTA, UNDP, Google.org humanitarian tech grants	Year 1–2

Table 3:Revenue Model

6.4 Pricing Strategy

- Government license: LKR 18–25 Million per year for national DMC deployment (all 25 districts)
- Municipal SaaS: LKR 500,000–1,500,000 per district per year
- Implementation and training: LKR 500,000–1,000,000 per agency engagement
- Pricing positioned well below equivalent international systems while reflecting the life-safety value delivered

6.5 Competitive Advantage

- Only system in the world supporting Sinhala/Tamil emergency NLP — no direct competitor exists
- Urgency-sensitive clustering is a novel architecture not present in any commercial CAD or dispatch system
- Built specifically for Sri Lankan disaster context — location gazetteer, dialect support, local incident types
- Academic credibility from published research strengthens government procurement confidence
- Modular architecture — C3 (SMS) can be deployed standalone for agencies not yet ready for full deployment

6.6 Intellectual Property Considerations

- Register software copyright with National Intellectual Property Office (NIPO) Sri Lanka — Month 3–4
- File provisional patent for the urgency-sensitive multilingual incident fusion pipeline — Month 7–8
- Publish research findings in peer-reviewed venues to establish academic priority

- Maintain the Sri Lankan emergency speech dataset and location gazetteer as proprietary assets
- Consider open-sourcing base Sinhala/Tamil NLP utilities to build community and research credibility while protecting the full integrated system

7. BUDGET AND JUSTIFICATION

CHAPTER 7: BUDGET AND JUSTIFICATION

Total Budget Summary

Category	Total (LKR)
7.1 Software Licenses	60,000
7.2 Cloud Services	30,000
7.3 Commercialization & IP	20,000
7.4 Miscellaneous	10,000
TOTAL 1-YEAR PROJECT BUDGET	**LKR 120000**

Table 4: Total Budget Summery

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase / Timeline	Component	Responsible Member	Key Tasks	Workload Contribution
Month 1 – Month 2	Project Planning	All Members	Literature Review, Requirement Analysis, System Architecture Design, Dataset Planning	Shared
Month 3 – Month 5	Voice-Based Language Identification, Incident Understanding & Urgency Detection	Member 1	Audio Preprocessing, Speech-to-Text Transcription, Multilingual Language Detection,AI agent to reply back fast to sms Panic/Prosody Analysis, Incident Classification, Location Information, Urgency Detection, Generating Structured JSON Output	Core responsibility of Member 1
Month 3 – Month 5	Multilingual Emergency AI Clarification Agent	Member 2	AI Conversation Agent Design, Information Not Provided, Question Generation, Multilingual Conversation Support (Sinhala, Tamil, English), Panic-Aware Conversation,	Core responsibility of Member 2

			Handling Interruptions and Call Drops, Generating Structured Conversation Summaries	
Month 3 – Month 5	Low-Bandwidth Text/SMS Emergency Intelligence	Member 3	Language Detection for Short Messages, Normalization of Short Message Text, Handling Emojis and Abbreviations, Handling Romanized Sinhala and Tamil, Incident Information, Urgency, Generating Structured SMS Intelligence Output	Core responsibility of Member 3
Month 6 – Month 7	Multimodal Incident Fusion, Timeline Reconstruction & Dashboard	Member 4	Integration of output from components 1-3, incident clustering, duplicate report detection, conflict resolution, creation of a unified incident record, timeline reconstruction, confidence scoring, visualization using a dashboard	Core responsibility of Member 4

Month 8	System Integration	All Members	Integration of all four components, communication via API, testing data flow between modules	Shared
Month 9	System Testing & Evaluation	All Members	Model evaluation, testing, measuring, validation	Shared
Month 10	Dashboard Improvement	Member 4+ Team	Improvement in visualizations, usability testing, refinement	Shared with lead Member 4
Month 11	System Optimization	All Members	Optimization, bug fixes, testing	Shared
Month 12	Documentation & Presentation	All Members	Writing final report, documentation, research paper, final presentation	Shared

Table 5: Work Breakdown Structure

9. GANTT CHART

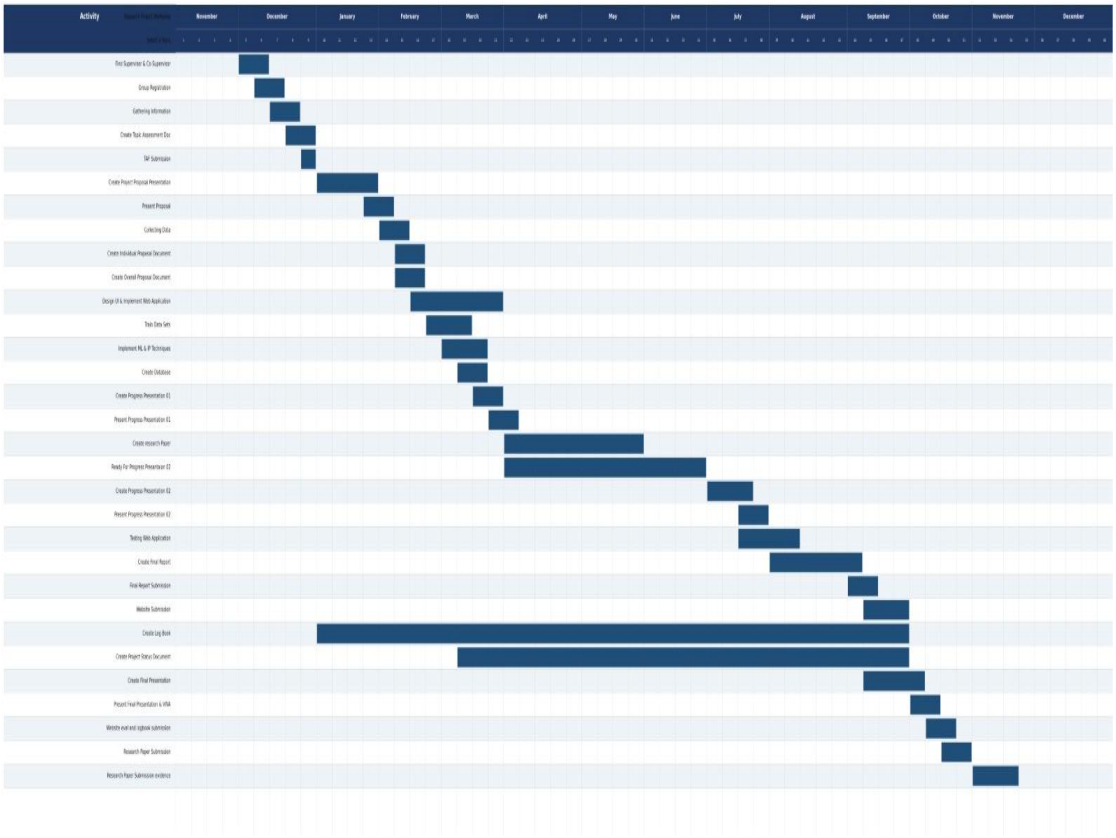


Figure 3:Gantt Chart

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11.APPENDICES

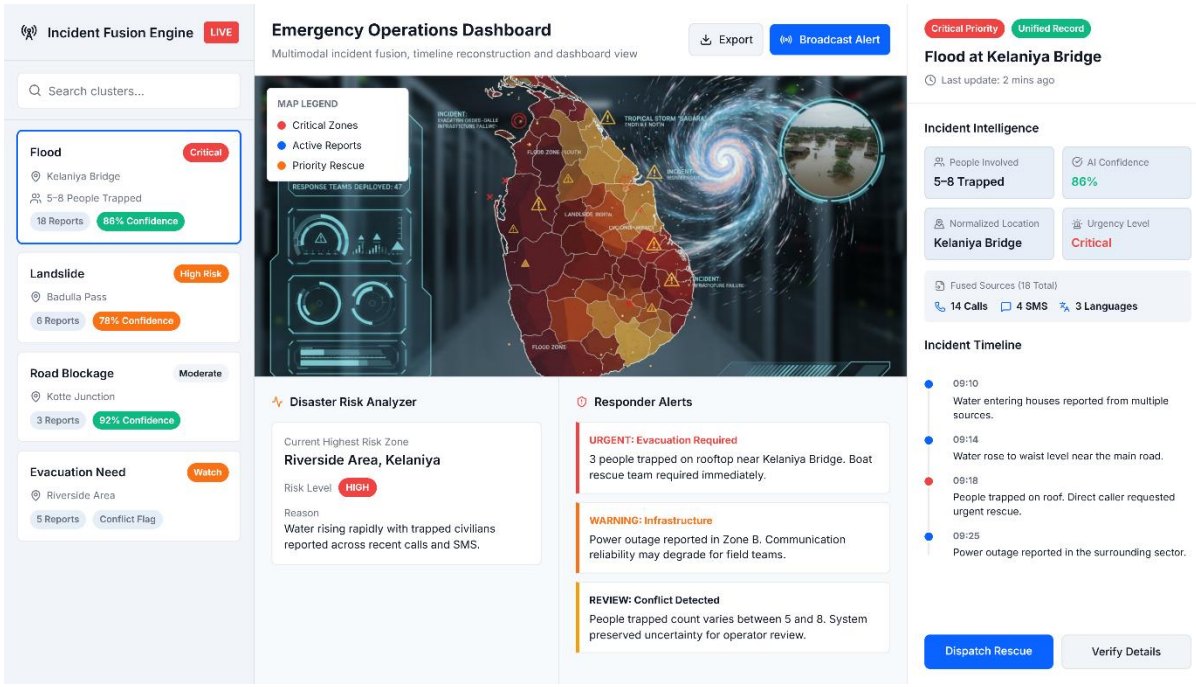


Figure 4-Ui mockup

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